THE LOCUS WHERE ALL COHOMOLOGY SHEAVES OF A PERFECT COMPLEX ARE FINITE FREE IS OPEN AND IS THE MAXIMAL SUCH OPEN SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- A finitely presented module with free stalk is free on a neighbourhood. Stacks Project, Modules on Sites, Lemma modules-lemma-finite-presentation-stalk-free (tag 0B8J). Spreads freeness of the stalk of the top cohomology sheaf to an open neighbourhood.
- Lifting a direct sum decomposition of a quotient. Stacks Project, Cohomology of Sheaves, Lemma lemma-local-lift-map (tag 08C6). Produces the direct sum decompositions of the strictly perfect complex used in the induction step.

Source reference. Original reference 0B8J: sources/stacks-chow/tags.txt, source lines 14142–14142. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08C6: sources/stacks-chow/tags.txt, source lines 10591–10591. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let E in $D(\mathcal{O}_X)$ be perfect. Assume that all stalks $\mathcal{O}_{X,x}$ are local rings. Then the set*

$$U = \{x \in X \mid H^i(E)_x \text{ is a finite free } \mathcal{O}_{X,x}\text{-module for all } i \in \mathbf{Z}\}$$

is open in X and is the maximal open set $U \subset X$ such that $H^i(E)|_U$ is finite locally free for all $i \in \mathbf{Z}$.

Proof. Note that if $V \subset X$ is some open such that $H^i(E)|_V$ is finite locally free for all $i \in \mathbf{Z}$ then $V \subset U$. Let $x \in U$. We will show that an open neighbourhood of x is contained in U and that $H^i(E)$ is finite locally free on this neighbourhood for all i. This will finish the proof. During the proof we may (finitely many times) replace X by an open neighbourhood of x. Hence we may assume E is represented by a strictly perfect complex $\mathcal{E}^\bullet$. Say $\mathcal{E}^i = 0$ for $i \notin [a, b]$. We will prove the result by induction on $b - a$. The module $H^b(E) = \mathrm{Coker}(d^{b-1} : \mathcal{E}^{b-1} \to \mathcal{E}^b)$ is of finite presentation. Since $H^b(E)_x$ is finite free, we conclude $H^b(E)$ is finite free in an open neighbourhood of x by Modules, Lemma 0B8J. Thus after replacing X by a (possibly smaller) open neighbourhood we may assume we have a direct sum

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decomposition $\mathcal{E}^b = \mathrm{Im}(d^{b-1}) \oplus H^b(E)$ and $H^b(E)$ is finite free, see Lemma 08C6. Doing the same argument again, we see that we may assume $\mathcal{E}^{b-1} = \mathrm{Ker}(d^{b-1}) \oplus \mathrm{Im}(d^{b-1})$. The complex $\mathcal{E}^a \to \ldots \to \mathcal{E}^{b-2} \to \mathrm{Ker}(d^{b-1})$ is a strictly perfect complex representing a perfect object E' with $H^i(E) = H^i(E')$ for $i \neq b$. Hence we conclude by our induction hypothesis. $\square$